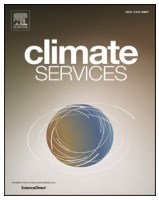
Climate Services 37 (2025) 100530



Contents lists available at ScienceDirect

Climate Services

journal homepage: www.elsevier.com/locate/cliser





Developing national frameworks for climate services: Experiences, challenges and learnings from across Europe

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ARTICLE INFO

Keywords
User-driven
Decision making
Risk management
Adaptation
Resilience
Climate services
Governance
National framework
Scenarios

ABSTRACT

Many countries are actively developing a National Framework for Climate Services (NFCS), driven by a need for better coordination of climate services to support decision making, mandatory climate adaptation and financial reporting, and the assessment of the quality of climate services including their underlying data. A group of five European countries (UK, Ireland, Belgium, Switzerland and Norway) has formed an informal peer-group to support, challenge, and share learning throughout the process of developing and evolving their NFCSs. This perspective article gathers these experiences to distil common themes and challenges as well as to identify unique characteristics and benefits of each country's experience on their NFCS implementation. While this article focuses on the experiences and practical learning within Europe, the benefit of establishing an NFCS is global, and many of the challenges and learnings presented here will be relevant anywhere in the world.

It is the experience of this group that peer support has accelerated and enhanced the NFCS implementation process, as well as provided a forum for exchange on how to maintain and evolve an operational NFCS. Based on this experience, we propose an additional step as part of the WMO's 'Step-by-step Guidelines for Establishing a National Framework for Climate Services' (WMO, 2018) to support countries to maintain and evolve their operational frameworks. Additionally, we recommend regional or global peer support on this framework development process should be sought, encouraged and facilitated.

Introduction

The World Meteorological Organization (WMO) recommends a National Framework for Climate Services (NFCS) as a national mechanism for coordination, governance and collaboration to improve the development, delivery and use of climate services at country level to support decision-making. Many countries are actively developing an NFCS, driven by a general need for better coordination of climate services. These services support research, investment and practice, as well as emerging needs such as mandatory adaptation and financial reporting, the growth of the climate services market and the need for assessment of quality and standards of climate services and their underlying data. Whilst there are useful guidelines from WMO along with some documented evidence of benefit provided by NFCSs around the world (e.g., Hewitt et al., 2020), there is a lack of literature sharing experiences and learning from the implementation process. This leaves countries starting out on this process without tangible examples to follow and tailor to their own context. There is also limited literature detailing the maintenance and sustainability of such frameworks, which would be of benefit to countries with more established and mature NFCSs.

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https://doi.org/10.1016/j.cliser.2024.100530
Received 26 January 2024; Received in revised form 20 November 2024; Accepted 21 November 2024
Available online 30 November 2024
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To address this challenge, five European countries, the United Kingdom, Ireland, Belgium, Switzerland and Norway have convened an informal peer-group to support, challenge, and share learning (Table 1). Each country has context-specific requirements for such a framework, and each has developed their NFCS through different channels and activities to meet that requirement. This article follows the journeys of the five countries through the lens of the '5 steps' of NFCS development and implementation (WMO, 2018) using a country case study for each step (Section 2). Learnings and challenges are explored (Section 3) in the hope that they might benefit other countries developing or maintaining an NFCS. A recommendation for an additional step to the WMO NFCS guidelines to support the maintenance and evolution of operational NFCSs along with final reflections is shared in Section 4.

Examining the process of developing an NFCS

The World Meteorological Organisation's 'Step-by-step Guidelines for Establishing a National Framework for Climate Services' (WMO, 2018) outlines '5 steps' of good practice for developing or establishing an NFCS (Fig. 1). While the European peer-group countries did not rigorously follow these steps, (many implemented their NFCS prior to the guidelines being published) on reflection the process taken by each country bore close resemblance to them. Therefore, it is a useful framework with which to examine the experiences of each country and to extract learnings that will be of relevance to other countries.

Establishing a baseline – illustration from Ireland

Ireland's 'project first' approach to establishing a baseline for their NFCS is a useful example which others have sought to follow and demonstrates a low-risk approach to securing stakeholder buy-in and further investment.

Following the Paris Agreement, Ireland introduced a National Adaptation Framework (DCCAE, 2018) requiring all government sectors to publish sectoral adaptation plans in 2019. All 31 local authorities in Ireland also developed sub-national regional adaptation plans. Along with changes in obligatory financial reporting of climate risk from the private sector, this created an unprecedented demand for climate information while also unintentionally creating confusion and duplication, with many more users, of various skill and knowledge levels, looking to obtain climate projections. At the time, Ireland had many climate service products in the public domain, developed by different stakeholders, with different data for different needs, but no standardised set of national climate projections. Following a review of the adaptation planning process, Ireland's Climate Change Advisory Council recommended that climate projections should be better coordinated. Establishing an NFCS was deemed an appropriate way to achieve this as well as wider coordination of national climate service delivery.

The country's state of readiness to implement an NFCS was assessed, and three clear opportunities emerged: (a) development of a national standardised set of climate projections, (b) coordinated co-development of standardised national climate services for multiple uses, and (c) improved accessibility and usability of climate services along with advisory support. Met Éireann, the Irish Meteorological Service, launched the research call 'TRANSLATE' (TRANSLATE, 2023), bringing together a critical mass of climate service stakeholders all working to provide standardised climate projections and climate services for Ireland (O'Brien and Nolan, 2023). The baseline assessment also identified (i) climate services currently being delivered; (ii) significant gaps; (iii) the primary providers and users of climate services and (iv) how national climate policy was likely to change these. The project developed case studies to illustrate the advantages of using standardised national climate projections to deliver climate services with cross-sectoral applications.

This 'project first' approach demonstrated that a permanent framework to coordinate provision of standardised climate information, often in a tailored format and to support increasing national demand both at a sectoral and cross-sectoral level was not only required but was in demand. An NFCS was ideally suited to help identify, support and streamline provision of national climate information and data. The appropriate stakeholders were already engaged due to their involvement in the TRANSLATE project. The next step was further national consultation to determine how best to introduce, develop and maintain an NFCS.

Table 1
A brief comparison of the five NFCS discussed in this article.

NFCS	Launched	Budget/Size	Brief description	Website
Norway	2011	400 K€ in dedicated funds, and almost 800 K€ in-kind from partners	The Norwegian Centre for Climate Services (KSS) is a virtual collaboration between the Norwegian Meteorological Institute, the Norwegian Water Resources and Energy Directorate, the Norwegian Mapping Authority, NORCE and the Bjerknes Centre. It aims to provide the physical scientific basis for climate change adaptation in Norway.	https://klimaservicesenter.no
United Kingdom	2013	Approx. 200 staff employed by the Met Office Hadley Centre for Climate Science and Services as well as in kind contributions	Climate Service UK formally launched a framework for UK climate services. Following recent consultation and evaluation a voluntary virtual 'Climate Services Hub' group facilitated by the UK Met Office focuses on building community, sharing knowledge and resources, and prioritising research activities such as Standards for Climate Services.	https://www.metoffice.gov.uk/services/research-consulting/climate-service
Switzerland	2015	1 FTE as executive director with large in-kind contributions from member institutions	The National Centre for Climate Services (NCCS) is the federal network for climate services. Organised in the form of a virtual centre, it brings together administrative federal bodies and federal research institutions to coordinate the collaborative development and provision of climate services.	https://www.nccs.ch
Ireland	2022	4 FTE with support from Met Éireann staff and in-kind contributions from member institutions	The Ireland NFCS is a virtual centre led by Met Éireann to support the coordination, development & provision of standardised national climate services. Ireland's existing climate adaptation portal 'Climate Ireland' helps disseminate adaptation related NFCS services.	https://www.met.ie/nfcs/
Belgium	2022	2 million euro annual budget	The Belgian Climate Centre is a federal scientific structure with the mission to maximise the impact of scientific research on climate mitigation and adaptation. Organised in the form of a physical centre where the administrative support is provided by the NMHS, the Royal Meteorological Institute of Belgium.	https://www.climatecentre.be

Undertake national consultation – illustration from United Kingdom

The UK experience of undertaking extensive national consultation within a well-developed but fragmented climate services landscape demonstrates the importance of engaging relevant stakeholders and jointly identifying what would be of most benefit with regards to governance, structure, investment and new research and coordination activities.

The UK has effective elements of an NFCS, as well as a growing number of academic, government, and non-governmental providers of climate services, many of these related to the provision of UK Climate Projections.[1] Climate Service UK was launched in 2013 to respond to the need for an NFCS.[2] In 2020, the UK Climate Resilience Programme funded national consultation to understand the need for further coordination. This initially took the form of bilateral discussions between the Met Office as the UK NMHS, and key relevant government departments. This established support and set priorities and boundaries for wider consultation. Six identical two-hour workshops were held with a broad cross-section of the UK climate services community, aiming to reach as many different actors as possible and provide opportunities to feed into discussion. This focused on determining the need for and benefits of further developing the NFCS, as well as what form this might take. 82 participants from 64 organisations across the UK joined the workshops and contributed views via interactive online boards. Workshops were held on different days of the week (thereby more inclusive) and technology was used to make the workshops strongly participatory. Participants were provided with a brochure ahead of the workshops describing the project and aims, and during the workshops participants were introduced to examples of NFCSs from other countries.

Perceived benefits of an improved NFCS were coordination, facilitation and strengthened collaboration among the UK climate services community and improved co-production, tailoring, delivery and use of science-based climate information and services. Participants expressed a clear desire for the government to play a central role, supporting the NFCS with resources, conferring legitimacy and credibility, essential for buy-in from the rest of the community. Participants commented on the need for initiatives raising awareness of the climate services landscape, greater knowledge exchange, and enhanced pull-through of climate services into adaptation action. Participants also raised ethical considerations around the role and remit of a national framework (such as a strong public good focus). Using this approach, recommendations were formulated for each theme according to evidence gathered in the workshops and organised into a broader recommendation for the development of the UK NFCS (Wilson et al., 2022).

This approach benefitted from the broad range of stakeholders able to have a 'voice at the table', and the inclusion of additional disciplinary and sector expertise. Inevitably it was not possible to reach all relevant actors, and future activities will focus on those more marginal to the climate services community but who hold key expertise or are critical to the delivery of climate services to vulnerable groups. Further benefits were establishing community participation and ownership of the process which provides momentum and engagement moving forward.

Develop an NFCS plan – illustration from Norway

The Norwegian Centre for Climate Services (NCCS-NO) exemplifies the importance of a clear and shared strategic plan when there are multiple partners involved, and it is useful to consider the process undertaken to achieve this. NCCS-NO was established in 2011 to provide a common national knowledge base for climate adaptation. Initially a cooperation between the Norwegian Meteorological Institute and the Norwegian Water resources and Energy directorate (NVE), it now also includes NORCE Norwegian Research Centre, The Bjerknes Centre for Climate Research, and the Norwegian Mapping Authority, ensuring collaboration between the main actors of climate research in Norway and avoiding possible overlap. Initial financing was based on in kind contributions and later a combination of direct contributions from the Ministry of Climate and Environment (KLD) and grants from the Norwegian Environment Agency. Due to limited financial resources early in its history, NCCS-NO's strategic plan has been a recent activity, and has proven important in driving the future direction and in seeking more sustained funding.

Municipalities and government agencies were involved in the development of the plan through projects, municipal networks and through the representation of the Environment Agency in the NCCS-NO steering committee and as commissioner of the knowledge base. Larger owners of infrastructure and sector authorities were involved as partners in projects (mostly financed by the Research Council of Norway) and as representatives in a national cross departmental group on climate adaptation hosted by the Environment Agency.

Under the vision "Knowledge for a climate robust society", the resulting strategic plan for the period 2020–2024 outlined the following goals:

1. NCCS-NO provides climate and hydrological data for the past and future in Norway. The data must be of high quality and available in a format that can easily be used.
2. NCCS-NO is the preferred supplier of knowledge about changes in climate and hydrology, and the effects of climate change on natural hazards.
3. NCCS-NO is one of Europe's leading national centres for knowledge about climate and hydrology for climate adaptation and impact research.

The strategic plan is closely aligned with the national strategy for climate adaptation (Meld. St. 26 (2022–2023)), and it includes the development of a national risk and vulnerability assessment (RVA) every fourth year, the first in 2026. NCCS-NO aims to launch updated national climate projections and an assessment report on climate change in Norway in 2025, thus providing input to the national RVA. In a parallel project, service design (for instance Dam and Teo, 2022) will be applied to improve the NCCS-NO service, which will contribute to a new strategic plan beyond 2024. The experience of NCCS-NO demonstrates the value of continuously evolving and revisiting elements of the NFCS implementation process, which should not be assumed to be unidirectional or linear and should be closely aligned with any strategic plans of individual partners or related programmes.

Endorse the plan and mobilise resource – illustration from Belgium

The experience of the Belgian Climate Centre has been useful within the peer group regarding the importance of gaining a mandate for the activities and how to achieve this, alongside wider stakeholder endorsement and resource mobilisation. In Belgium, climate research expertise is spread between universities, regional research institutes and federal scientific institutes. Baseline assessments highlighted the need for a structural framework for climate research and services with partnerships between different regions and communities, both at the policy and research levels (Brasseur et al., 2015; Fonteyn, 2015). The National Adaptation Plan (2017–2020) also specified the need for high-resolution national climate scenarios and development of a road map for the creation of a Belgian Climate Centre.

In 2021, the Secretary of State for Recovery and Strategic Investments formed a working group represented by three Federal Scientific Institutes (including the Belgian NMHS), the Federal Public Planning Service Science Policy (BELSPO), and a steering group with

[1] https://www.metoffice.gov.uk/research/approach/collaboration/ukcp/about/what-is-ukcp.

[2] https://www.metoffice.gov.uk/services/research-consulting/climate-service/climate-service-uk.

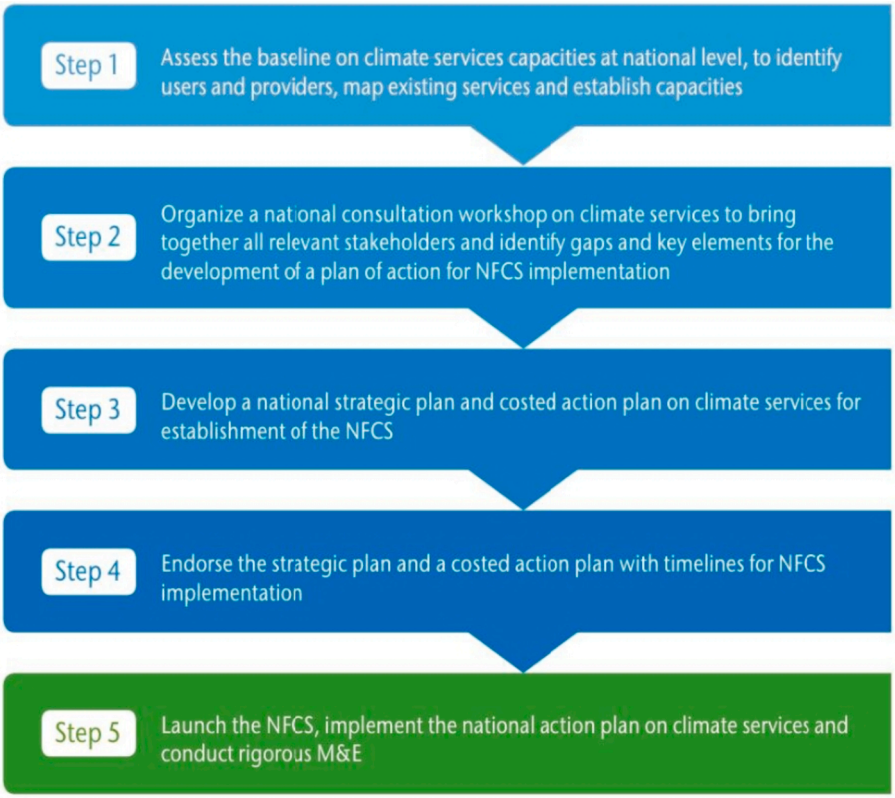


Fig. 1. Five steps for establishing an NFCS (WMO, 2018).

representatives from international, federal and regional research institutes and the private sector. This group was given the mandate to plan for the creation of a Belgian Climate Centre, which resulted in governmental agreement and an annual budget of EUR 2 million per year. This Centre, organized in the form of a physical centre, was launched in 2022 with objectives to: (i) bundle and strengthen climate-related research capacity by establishing structural collaborations with research institutes at the federal and regional level as well as with universities, hereby putting forward complementarity and interdisciplinarity as key values, (ii) improve the transfer of knowledge from researchers to users by developing climate services in order to enable science-based decision making, by facilitating the access to data and information, connecting experts and users and performing expert analyses and consulting, and (iii) increase the coherence and strategic impact of research programs.

To confer added legitimacy and sustainability to the activities, the federal Parliament adopted a climate law in December 2023 anchoring the establishment of a committee of independent experts to ensure a proper assessment of federal climate policy, the Secretariat of which is hosted at the Belgian Climate Centre.

Implement and evaluate the NFCS – illustration from Switzerland

One of the most well established NFCSs within the peer group, Switzerland provides an example to other countries in the importance of ongoing maintenance and evolution of structures, governance and activities to ensure the NFCS continues to meet its purpose and aims. The NFCS in Switzerland has been operational since 2015, when the National Centre for Climate Services (NCCS-CH, 2015) was founded as a virtual centre bringing together the Federal Offices of the environment, agriculture, civil protection and meteorology/climatology, and two Federal research institutions (ETH Zurich and WSL), mandated through the National Climate Adaptation Strategy. As a federal network, the NCCS-CH functions as an interface between producers and users, promoting dialogue and the collaborative, user-centred development of climate services.

Since its foundation in 2015 NCCS-CH has substantially evolved. First, additional Federal Offices joined the network as new members (Federal Offices of health, animal health and energy). Second, major projects in the form of priority themes, each put forward by a NCCS-CH member, have been undertaken. Priority themes are organised along the climate services value chain and include, among others, the national climate change scenarios CH2018 (Fischer et al., 2022; NCCS, 2018), hydrological scenarios (NCCS, 2021a), climate impacts on forest functions, crop pests and civil protection as well as the elaboration of a national hail climatology (NCCS, 2021b). Third, the website https://www.nccs.ch was launched in 2018 alongside the publication of the national climate scenarios, acting as a central information hub for users and producers of climate services. A fourth development concerns a recently launched programme on climate impacts (NCCS-Impacts, 2022-2026). For the first time, funding from policy research of the Federal Offices is pooled together with substantial in-kind contributions of all members. Transdisciplinary and diverse consortia seek to progress cross-sectoral topics including socioeconomic scenarios, health and animal health impacts, impacts on critical energy infrastructures, supply chains and ecosystem services, as well as an assessment on the costs of climate change in Switzerland (Fischer et al., 2024; Geiger et al., 2024).

The NCCS-CH is also being constantly monitored and evaluated. One central measure is the organisation of an annual forum bringing together the main stakeholders from research, administration and the private sector (around 200 participants). The aim is to discuss challenges with currently available climate services and to identify new needs and services. It is also a networking opportunity to foster dialogue among actors, bring in additional transdisciplinary expertise and to envisage new co-production projects and actionable services/products. The NCCS-CH is also monitored via bi-annual meetings with and reporting to the board of directors and executive committee. The combination of these events ensures that NCCS-CH is meeting its objectives whilst advancing to meet new needs.

Recommendation for a sixth step – review and evolve

At present there is no available guidance to support the ongoing maintenance and evolution of an NFCS once established, though this was considered critical by all members of the peer group. Establishing standards and encouraging best-practice approaches, as well as bringing the climate services community together in discussions around, for instance, codes of ethical practice, can foster the quality of climate services way beyond the inception of an NFCS. There is also a strong need to evidence the socio-economic benefits the NFCS is providing, and to demonstrate inclusivity and equity in activities, all of which require specific expertise and guidance. The NCCS-CH example exemplifies the need and opportunity to keep evolving after implementation to ensure needs continue to be met, and examples from the UK and Norway clearly demonstrate that steps such as national consultation and strategic planning should be revisited periodically. From the European experience, NFCS maintenance and evolution should involve two complementary pathways:

a. *Climate services in line with latest science and evolving societal needs*

The provision of national climate services, such as projections and associated products, must be maintained and evaluated to keep up with the latest science, international climate projections and evolving user demands. This involves the maintenance and development of existing and new services tailored to defined needs. New perspectives on evolving climate science and policy development across disciplines as well as temporal and spatial scales from global, continental, national or sub national scales, require continuous support from the NFCS to translate and contextualise new developments to users. This is particularly important as we look to a future of digital twins, artificial intelligence and machine learning.

b. *Sustain and broaden stakeholder interaction*

Once an NFCS is operational, continuity in established collaborations with stakeholders is crucial and requires evaluation and possible revision of the user interface and communication methods. The communication of updated climate services based on new projections or scenarios can be challenging. Services and communication products must also be adapted and expanded based on new stakeholder needs, reaching different sectors (e.g. industry, tourism) or the involvement of other transdisciplinary partners or knowledge holders.

This stage is challenging and the experience of the European countries shared in this perspective is that this requires additional inputs from a range of disciplines as well as across NFCS stakeholders to ensure success. One notable challenge is the requirement to monitor and evaluate ongoing progress and benefits of the NFCS, particularly if an increase in funding and resourcing is required. Whilst a range of examples and guidance exist to demonstrate the socio-economic benefits of weather and climate services on an individual basis, or for a particular sector (see for instance Perrels et al., 2013; Frei, 2010; Bruno Soares et al., 2018; WMO, 2015) there is limited quantifiable or evidence-based information available on the benefits of the NFCS, or how to go about assessing this, and this guidance is urgently needed.

Building on the experiences within the European peer support group, it is recommended to include an additional Step 6 in the Step-by-Step guidance for establishing an NFCS which supports maintaining and developing an NFCS on a permanent status by embedding structural and long-term sustainability. Including this emphasis and additional guidance on a Step 6 will help to ensure that the value of the NFCS for each country is maximised and maintained.

Discussion – learnings & challenges

There are many formal European forums where climate services and coordination frameworks, projects and initiatives are discussed (e.g. EUMETNET, 2024; Copernicus, 2024; Climateurope2, 2024 and WMO region VI). However, informal peer support offers an environment where seemingly trivial questions can be asked and topics that are more controversial in nature can be explored. The informal non-minuted environment allows discussions on topics such as relative costs, unintended competition & potential liability to be safely explored. This perspective shares some of the learnings and challenges common to peer-group members.

Learnings

There is no 'one size fits all': No peer-group members started their NFCS from a blank page, with many countries already having pre-existing climate service coordination initiatives. Members found there is no predefined exact route in establishing an NFCS, as evidenced in Section 2. Rather, the national contexts widely differ and so do the required specifications. Countries will need to focus on different components of a climate services framework, according to national priorities, and these needs will evolve and change over time as national capacity and capability grows.

Government stakeholder support is critical: When trying to establish a national framework, having support of a major national stakeholder from the start is important (e.g. the NMHS, a national environmental agency, or major sector such as agriculture, water or energy). This provides credibility from an early stage and can help align existing climate coordination initiatives. A government mandate to establish an NFCS helps to increase coordination authority and legitimacy of the process.

Usability and tailored content are essential for wider reach: Regular, scheduled feedback on existing climate services and associated guidance are essential. Prolonged interaction with key user groups is required to ensure delivery of successful services. There are also gains from disseminating climate services in a standardised way. This not only promotes the service but also provides the opportunity for cross-sectoral benefits.

Openness and transparency bring benefits: An open, transparent and easy to engage NFCS can dramatically improve the availability and uptake of climate data and information. This reduces duplication of effort and the emergence of similar, but different, climate products. This open usability reduces entry barriers to commercial climate service providers, promoting job creation as well as accelerating the spin up of research projects.

Regional or international exchange is helpful: An international exchange on country-wise implementation of an NFCS can be highly beneficial. Any country either starting to establish or looking to improve their NFCS should consider forming such a peer-group. Although having different contexts, there are typically similarities related to research, dissemination of services, user engagement, policy advice, private sector relationships and transboundary considerations (such as water catchments) across neighbouring or nearby countries.

Challenges

Governance: Establishing a functioning NFCS governance committee where each member has equal say and decisions are taken unanimously can be challenging. Often the organisation mandated to lead the NFCS is also a participating member. This has the potential to lead to governance difficulties, however this mandate also provides useful authority in leadership and facilitation.

Resourcing: Sustainable resourcing is needed to maintain an NFCS and support development of climate services. This has been a significant constraint on peer group members at times. Supporting the development of continuous climate services to align with climate adaptation planning cycles and mitigation targets also requires a sustained commitment.

Unrealistic expectations: Managing the expectations of what an

NFCS will do, and not do can be challenging. Finding the lines between where the development of a national climate service stops and where the handover to a user or researcher begins, is not always clear. Also clearly defining what an NFCS is supporting and what an organisation within the framework would be doing anyway can cause confusion of ownership or recognition.

Complex landscapes: The climate services landscape and associated ecosystems can be broad and with a wide range of actors with their own priorities and working practices. It is common for duplication and overlap to occur and therefore some level of conflict or disagreement may arise. Supporting research and job creation versus the long-term, open and sustainable provision of national climate services is a difficulty facing most NFCSs. Clear branding of an NFCS is useful but can be difficult as typically these frameworks are supported by multiple national institutions, each with their own accreditation and logos which they may want appropriately accredited in any coordinated outputs.

Ensuring quality and good practice: Quality assurance of climate services, tension between better data versus better decisions (cf. Findlater et al., 2021) and cost-benefit estimations (Perrels, 2020) is an issue for any climate service provider and a challenge for an NFCS coordinating across the landscape.

These challenges were common to all or most of the countries discussed in this perspective article, and the sharing of experiences and dialogue around potential solutions has been of great value. For instance, the different routes to initial and sustained funding in each country have been discussed and the use of in-kind contributions from partners (e.g. Norway example) and the ongoing joint curation of large research projects (e.g. Swiss example) have proven effective models. Some of these challenges remain so, and it is here that the continued sharing of ideas and experiences, along with an updating of the WMO Step-by-Step Guidelines to provide targeted support is much needed.

Final reflections

Mapping the development of the five peer-group members from a heterogeneous starting point, with Norway's preceding the five steps framework, and Belgium and Ireland being the youngsters in the group, it is striking how each NFCS has gravitated towards a similar implementation with common benefits experienced. In retrospect the five steps can be recognised in the development of each NFCS, albeit not always in the same order. This showcases the value of the 5 Step Guidelines for any country anticipating NFCS implementation. However, the experiences shared among the five NFCSs in this perspective show that there is a large heterogeneity with very country-specific boundary conditions, which makes it difficult to directly intercompare the specific developments and setups of the different NFCS in a structured way. This highlights the importance of sharing learning between countries to appreciate what has worked well and could be well suited to different country contexts.

A key learning from the exchange of experiences is to include an additional sixth step within the WMO guidelines. This concerns a continuous review and evolve step that is required to steadily improve the coordination, governance and collaboration once the NFCS is in operation, ultimately to further the development, delivery and use of climate services. An important aspect of this sixth step could be to share learning and seek support from regional networks of NFCSs and wider regional activities. Such networks, as the participants in this perspective article represent, are an important tool to continuously improve the NFCS based on learning from others. The informal peer-group represented in this article is small enough that every voice is heard, but big enough to generate knowledge spread across the NFCS journey. It is the intention to continue this informal group into the future and we recommend consideration of establishing similar groups.

At times of accelerated need for trusted and robust climate information, provided in ways useful and usable by societal actors, this perspective article illustrates the advantages of establishing an NFCS with peer support. The article also highlights the need for a framework to evolve to meet stakeholders ever changing climate information needs. It is this reason that we recommend the inclusion of a sixth step within the WMO guidelines to support continuous NFCS evolution, to ensure access to coordinated authoritative and relevant climate services.

CRediT authorship contribution statement

Nicola Golding: Writing – review & editing, Writing – original draft. **Keith Lambkin:** Writing – review & editing, Writing – original draft. **Louise Wilson:** Writing – review & editing, Writing – original draft. **Rozemien De Troch:** Writing – review & editing, Writing – original draft. **Andreas Marc Fischer:** . **Hans Olav Hygen:** Writing – review & editing, Writing – original draft. **Angela Michiko Hama:** Writing – review & editing, Writing – original draft. **Anita Verpe Dyrrdal:** Writing – review & editing, Writing – original draft. **Ella Jamsin:** Writing – review & editing, Writing – original draft. **Piet Termonia:** Writing – review & editing, Writing – original draft. **Christopher Hewitt:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank everyone who we have collaborated with and helped us understand the climate service landscape. This article has been published with the support of the World Meteorological Organization (WMO). The opinions, findings, interpretations, and conclusions expressed in this article are those of the authors and do not purport to reflect the opinions of WMO or its Members. We also thank the WMO for permission to reproduce Fig. 1.

Data availability

No data was used for the research described in the article.

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